

6.1. DIRECT OR INDIRECT?

I concentrate my analysis on institutional direct real estate investment and indirect real estate investment through REITs. REITs are mutual fund-like vehicles allowing broad ownership of real estate assets. Many REITs are publicly traded. REITs have to satisfy many restrictions: most notably, they are required to distribute at least 90% of their taxable income each year to shareholders as dividends. REITs are one of the few ways retail investors can access commercial real estate markets and were created by Congress in 1960 through the Real Estate Investment Trust Act. Congress did not have the high-minded intention of allowing the masses access to the diversification benefits of commercial real estate; the REIT Act was originally a rider attached to An Act to Amend the Internal Revenue Code with Respect to the Excise Tax on Cigars and was passed after adept lobbying by the real estate industry so that developers could tap the equity market for additional funds.³¹

Since direct real estate and REITs both involve buildings and land that generate cash flows, one would think the returns of these two approaches would be highly correlated.³² Surely direct and indirect physical holdings of real estate must be driven by common fundamentals in the long run. But they are not, and this is called the *REIT puzzle*.

Table 11.11 reports means, standard deviations, autocorrelations, and cross-correlations of direct and indirect real estate returns along with equity returns from the beginning of 1978 to the end of 2011 at the quarterly frequency. The direct property returns are from the National Council of Real Estate Investment

Table 11.11

	Mean	Stdev	Autocorrelation
NCREIF	0.0221	0.0225	0.7806
REIT	0.0347	0.0918	0.1070
S&P 500	0.0301	0.0823	0.0711
Correlation Matrix			
	NCREIF	REIT	S&P 500
NCREIF	1.0000	0.1520	0.0900
REIT		1.0000	0.6265
S&P 500			1.0000

³¹ See Brandon (1998) and Graff (2001).
³² For a theoretical basis of this proposition, see Carlson, Titman, and Tiu (2010).